CCC-C-447c
February 19, 1968
SUPERSEDING
Fed. Spec. CCC-C-447b
June 6, 1963

FEDERAL SPECIFICATION

CLOTH, PERCALE, COTTON

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers one type of cotton percale cloth.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards of the issues in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

Federal Specifications:

CCC-T-191 - Textile Test Methods.
PPP-P-1134 - Packaging and Packing of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics).

Federal Standards:

Fed. Std. No. 4 - Glossary of Fabric Imperfections.
Fed. Std. No. 595-Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards, and at the prices indicated in the Index The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids or request for proposal shall apply.

Standard Color Card of America.

FSC 8305

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(Copies may be obtained from The Color Association of the United States, Inc , 200 Madison Avenue, New York, N Y. 10016.)

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained free of charge from the Federal Trade Commission, Washington, D C. 20580)

3 REQUIREMENTS

3 1 Standard sample. When a standard sample is available, the dyed, printed, or bleached cloth shall match the standard sample for shade and design when applicable and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6 3).

3.2 Material.

3.2.1 Yarn. The yarn shall be made from cotton which has been cleaned, carded, drawn and spun into single yarn for both the warp and the filling.

3.3 Color. The color shall be bleached white, dyed, or printed as specified (see 6.2). The use of sulfur dyes or dyes containing elementary sulfur compounds capable of oxidation to sulfuric acid is prohibited. The dyestuffs shall be chosen and applied so that the dyed or printed cloth shall show no more labile sulfur than shown by the standard sample when tested as specified in 4 4. When no standard sample is established there shall be a minimum of staining of the test paper when tested as specified in 4.4. The dyed colors shall match the following numbers of Fed. Std. No 595 and cable numbers of the Standard Color Card of America.

Fed. Std. No 595		Standard Color Card of America	
Color	Number	Color	Cable Number
Green	24260	Blue	70023
Purple blue	35231	Brown	70107
Tan	30277	Light blue	70021
White	37875	Red	70080
Yellow	33695		

3.3.1 Matching. When a standard sample is available, the dyed or printed cloth shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3 3 2 Colorfastness. The dyed or printed and finished cloth shall show colorfastness to wet dry cleaning, crocking combined laundering and bleaching, perspiration, and light equal to or better than the standard sample. When no standard sample is established, the dyed or printed and finished cloth shall show "good" colorfastness to wet dry cleaning, combined laundering and bleaching, perspiration and light, and to crocking not lower than Munsell value 8 5 (except that for Black or Dark Blue cloth, colorfastness to crocking shall be not lower than Munsell value 6 5) when tested as specified in 4.4

3.3 3 Printed cloth When printed cloth is required, it shall be in color patterns and designs of floral, geometric, striped plaids or checks, as specified (see 6.2).

3 4 Physical requirements. The finished cloth shall conform to the requirements specified in table I when tested as specified in 4 4

3 4 1 Width The width shall be as specified (see 6.2) and shall be the minimum acceptable width inclusive of selvages

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3.5 Finish. The cloth shall be singed, bleached, printed, or dyed and may be calendered. The starch and protein content including chloroform-soluble and water-soluble material shall not exceed 2.5 percent when tested as specified in 4.4.

3.6 Acidity (pH). The pH value of the water extract of dyed and finished cloth shall be not less than 5 5 nor more than 8.5 when tested as specified in 4 4.

3.7 Shrinkage. The cloth shall be preshrunk and the residual shrinkage shall not exceed 2.0 percent in either the direction of the warp or the filling when tested as specified in 4.4. The preshrinking process used shall not be identified on the cloth, the ticket, or on the package.

3.8 Length and put-up. Unless otherwise specified (see 6.2), the finished cloth shall be furnished in continuous lengths, each not less than 40 yards. Each length shall be put up in flat folds as specified in PPP-P-1134.

3.9 Fiber identification. Each piece shall be labeled, ticketed, or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

3.10 Workmanship. The cloth shall conform to the quality and grade of product established by this specification. The demerit points per 100 square yards when calculated as specified in section 4 shall not exceed the established maximum point value.

TABLE I. Physical requirements

Weight per square yard (minimum)	Yarns per inch (minimum)		Breaking strength (minimum)		Weave
	Warp	Filling	Warp	Filling	
Ounces			Pounds	Pounds	
3.0	85	72	45	30	Plain

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105 except when otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 Examination of the end item. Examination of the end item shall be in accordance with 4.2.2.1 through 4.2.2.3.

4.2.2.1 Yard-by-yard examination. The entire yardage of each piece in the sample but not more than 100 yards per piece, shall be examined on the face side. Defects as defined in Section 1 of Fed. Std. No. 4 which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in

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4 2.2.1.1 except that only those slubs and knots which exceed the maximum limits shown in figure 1 shall be scored. No folded linear yard shall be penalized more than 4 points. The sample size shall be 20 pieces selected from 20 containers. For determination of acceptability of the average quality of the lot, the lot shall be unacceptable if the points per 100 square yards exceed the following values·

30 0 points for dyed or printed cloth
35.0 points for white cloth.

For determination of acceptability on the individual piece quality basis, the lot shall be unacceptable if two or more individual pieces exceed the following values

45.0 points for dyed or printed cloth
53.0 points for white cloth.

If no individual piece exceeds the point level, the lot shall be accepted. If one piece exceeds the point level, a second sample of 20 pieces shall be selected for individual piece quality examination only. If no piece in the second sample exceeds the point value, the lot shall be accepted. If one or more pieces exceeds the value, the lot shall be rejected. For individual piece quality examination, local type defects only shall be scored. Overall type defects shall be scored only in determining compliance with the lot average quality requirement. Point computation for lot quality and individual piece quality shall be as follows·

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches} \times \text{total yards inspected)}} = \text{Point per 100 square yards}$$

4.2.2 1 1 Demerit points Demerit points shall be assigned as follows:

For defects up to 3 inches in any dimension	-	one point
For defects exceeding 3 inches but not exceeding 6inches in any dimension	-	two points
For defects exceeding 6 inches but not exceeding 9 inches in any dimension	-	three points
For defects exceeding 9 inches in any dimension	-	four points

The following defects, when present, shall be scored four points for each yard in which they occur:

Baggy, ridgy, or wavy cloth.
Objectionable odor
Width less than specified.
Poor dye penetration, mottled, streaky, or cloudy.
Excessive neppiness
Prints consisting of multiple colors, out of register
Prints not clearly defined.

4 2 2 2 Examination for length

4 2.2 2 1 Individual pieces During the yard-by-yard examination, each piece in the sample shall be examined for length Any length found to be less than the minimum specified or more than two yards less than the length marked on the ticket shall be considered a defect with respect to length The lot shall be unacceptable if two or more pieces in the sample are defective in respect to length.

4.2.2.2.2 Total yardage in sample The lot shall be unacceptable if the total of the actual lengths of pieces in the sample is less than the total of the lengths marked on the tickets

4.2.2.3 Examination for color and shade. During the yard-by-yard examination, each piece in the sample shall be examined for color, shade and color pattern and design. Any piece in the sample off shade, shaded side to side, side to center, end to end, or incorrect color pattern or design shall be cause for rejection of the entire lot represented by the sample

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4.2.2.4 **Examination for identification of preshrinkage process and noncompliance with Textile Fiber Products Identification Act.** During the yard-by-yard examination, each piece in the sample shall be examined for these defects. The lot shall be unacceptable if two or more pieces in the sample contain identification of the preshrinkage process by name or trademark on the cloth or ticket, or are not labeled, ticketed or invoiced in accordance with the Textile Fiber Products Identification Act.

4.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-1134 to determine that packaging, packing, and marking requirements of section 5 of this specification are complied with.

4.4 Testing of the end item. The methods of testing specified in CCC-T-191, wherever applicable and as listed in table II shall be followed. The physical and chemical values specified in section 3 except where otherwise specified, apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method. All test reports shall contain the individual values utilized in expressing the final result. The sample unit shall be 3-1/2 continuous yards, full width of the finished cloth. The lot size shall be expressed in units of 1 yard. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified. The sample size shall be as follows:

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE II. Tests

Characteristic	Requirement paragraph	Test method
Identification of cotton	3.2	1200 1/
Weave	3.4	Visual 2/
Weight, ounces per square yard	3.4	5041
Yarns per inch.		
Warp	3.4	5050
Filling	3.4	5050
Breaking strength:		
Warp	3.4	5100
Filling	3.4	5100
Sulfur content	3.3	2020
Colorfastness to:		
Light	3.3.2	5660
Combined laundering and bleaching	3.3.2	5605
Wet dry cleaning	3.3.2	5622
Crocking	3.3.2	5651
Perspiration	3.3.2	5680
Starch and protein content	3.5	2611
Acidity (pH)	3.6	2811
Shrinkage in laundering	3.7	5550

1/ A supplier shall submit a certificate of compliance for this characteristic. When required, the certificate shall be accompanied by actual test, inspection or other verifiable data.

2/ One determination shall be made from each sample unit and the results reported as "pass" or "fail".

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

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5.1.1 Levels A and C. The cloth put up as specified shall be packaged in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Levels A, B, and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1134.

6. NOTES

6.1 Intended use. The cotton percale cloth covered by this specification is intended for use in the manufacture of clothing.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Color of cloth required (see 3.3).
(c) Type of designs or patterns of printed cloth (see 3.3.3).
(d) Width of cloth required (see 3.4.1).
(e) When length of piece is other than specified (see 3.8).
(f) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample, address the procuring office issuing the invitation for bids.

6.4 Color (civil agencies). The colors designated in 3.3 are established to meet civil agency requirements for the following Federal stock numbers, as indicated:

Color	Federal Stock Numbers (FSN)	
Light Blue	Turquoise Blue	(8305-205-3066)
Brown	Dark Brown	(8305-205-3067)
Purple Blue	Copenhagen Blue	(8305-205-3068)
Blue	Peacock Blue	(8305-205-3069)
Tan	Light Tan	(8305-205-3070)
Green	Green	(8305-205-3071)
Red	Dark Red	(8305-205-3167)
White	White	(8305-205-3168)
Yellow	Yellow	(8305-205-3169)

MILITARY CUSTODIANS:

Army - GL
Navy - SA
Air Force - 82

Review activities:

Army - GL, MD
Navy - SA
Air Force - 82, 85

Preparing activity:

Army - GL

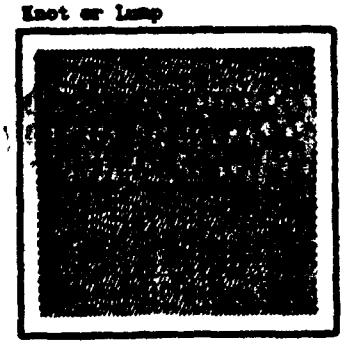
CIVIL AGENCIES INTEREST:

AGR
COM-NBS
DC
GSA
VA
HEW

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Limits of Acceptability

Knot or Lump



Yarn Thickening (Slub)

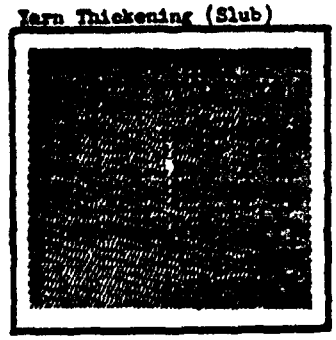


FIGURE 1.

Orders for this publication are to be placed with General Services Administration acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price of this specification 5 cents each.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION
Cloth, Percale, Cotton - CCC-C-447c

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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